Incidence Perfect NG

User Manual (Beta)

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Incidence Perfect NG is a compact 2-axis inclinometer/incidence meter built around an ESP32-S3 + AMOLED touchscreen. It measures **roll** and **pitch** and provides workflows for **ZERO**, **OFFSET CAL**, **MODE**, **ROTATE**, and **ALIGN**.

> Beta note: This manual describes the current firmware behavior. If something differs on your device, report the **firmware version shown on the splash screen**.

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1) Getting Started

Power + Boot

- Connect the device over USB power.
- On boot you'll see the splash screen with the firmware version in the lower-right.
- If the UI was previously set to `ROT 180`, the splash screen follows that stored orientation.
- If `Startup ZERO` is enabled, the device begins guided ZERO automatically on cold boot.
- After boot, the live readout screen appears (roll/pitch).

What You're Looking At

- **Top status line** shows:
  - orientation mode (`SCREEN UP` or `SCREEN VERTICAL`)
  - axis view (`BOTH`, `ROLL`, `PITCH`)
  - rotation (`ROT 0` or `ROT 180`)
  - live state (`LIVE` or `FROZEN`)
- **Top-right battery label** shows:
  - `BAT <percent>% <voltage> V`
  - `BAT? <percent>% <voltage> V` when firmware suspects USB rail without battery pack
  - a steady charge icon when charging is detected
- **Readouts**:
  - `ROLL` (left) and `PITCH` (right) in degrees
  - colors shift for large angles (warning, then critical)
- **Bottom buttons**:
  - `ZERO`, `AXIS`, `MODE`, `ALIGN`, `ROTATE`

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2) Touch Controls (Everyday Use)

ZERO / OFFSET CAL

Use when the tool is resting in the reference position.

- Tap `ZERO`.
- `ZERO` opens a guided workflow:
  - `CONFIRM` to start stillness + averaging
  - `CANCEL` to abort
- During apply, a progress bar is shown.
- On completion, values settle around `0.00`.
- Long-press `ZERO` to start guided `OFFSET CAL` (confirm/cancel workflow).

AXIS

Choose what you want to focus on:

- Tap `AXIS` to cycle: `BOTH -> ROLL -> PITCH -> BOTH`.

ROTATE (180 degrees)

Use when the device is physically hard to read and you want to flip the UI.

- Tap `ROTATE` to toggle `ROT 0` / `ROT 180`.
- Rotation persists after reboot.
- The boot splash follows the same stored rotation.

Freeze / Unfreeze

Use freeze when you want to capture a reading without chasing tiny motion.

- Tap the **readout area** (the roll/pitch values) to toggle `LIVE` / `FROZEN`.
- When frozen, the displayed values hold steady.

Touch Layouts: Advanced vs Simple

The device supports two touch layouts:

- **Advanced**: full button row (`ZERO`, `AXIS`, `MODE`, `ALIGN`, `ROTATE`)
- **Simple**: two large buttons (`ZERO`, `VIEW`)

`VIEW` in **Simple** combines axis and mode control:

- **Tap `VIEW`**: cycle axis scope
  - `BOTH -> ROLL -> PITCH -> BOTH`
- **Medium hold `VIEW` then release**: toggle `MODE`
  - `SCREEN UP <-> SCREEN VERTICAL`
- **Long hold `VIEW` then release**: switch touch layout
  - `Simple <-> Advanced`

In **Advanced**:

- **Tap `MODE`**: toggle orientation mode
- **Long hold `MODE` then release**: switch touch layout to `Simple`

Tip behavior:

- During hold, the tip line shows what action will happen on release.
- In `Simple`, `VIEW` tip bands are:
  - `AXIS` (short hold)
  - `MODE` (medium hold)
  - `SWITCH UI` (long hold)

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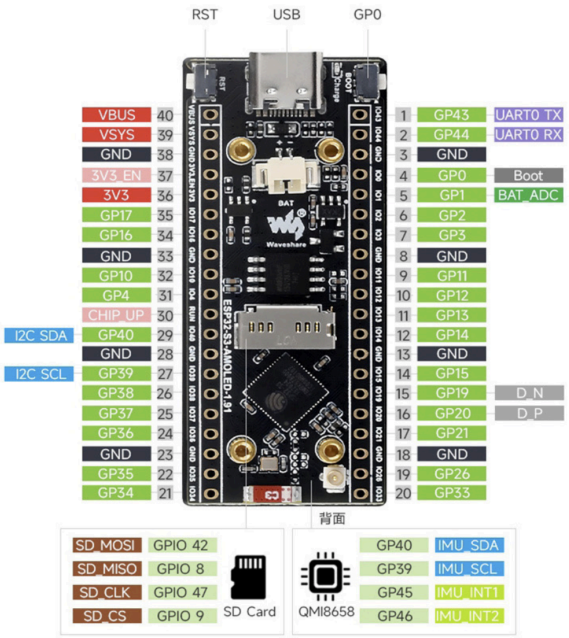
3) ACTION Button (Physical Control)

The board has a physical **ACTION button** (electrically `GPIO0`, labeled `BOOT/GPO` on the board). It mirrors key actions so you can operate the device when the screen is hard to reach.



In normal measurement mode:

- **Short press**: toggle freeze (`LIVE <-> FROZEN`)
- **Long press (~1.2s)**: cycle `AXIS` (`BOTH -> ROLL -> PITCH`)
- **Very long press (~2.2s)**: toggle `MODE` (`SCREEN UP <-> SCREEN VERTICAL`)
- **Ultra long press (~3.2s)**: start guided `OFFSET CAL` workflow
- **Super long press (~5.0s)**: enter deep sleep (press ACTION to wake)

While holding the ACTION button, an on-screen hint shows what will happen on release and a progress indicator for the next threshold. Countdowns are shown as `x.x s` (with a space before `s`).

In guided `OFFSET CAL` workflow:

- **Short press**: `CONFIRM`
- **Long press (~1.2s)**: `CANCEL`

In guided `ZERO` workflow:

- **Short press**: `CONFIRM`

- **Long press (~1.2s)**: `CANCEL`

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4) MODE (Orientation Change)

`MODE` changes how the device interprets orientation.

- `SCREEN UP`: standard “screen facing up” use case
- `SCREEN VERTICAL`: use case where the tool is used on a vertical surface

Touch Workflow

1. Tap `MODE`.
2. Orientation switches immediately between `SCREEN UP` and `SCREEN VERTICAL`.
3. Mode-specific stored calibration/zero references are loaded automatically.

Serial / Web

- Serial:
  - `m`: toggle mode
  - `u`: set `SCREEN UP`
  - `v`: set `SCREEN VERTICAL`
- Web:
  - `MODE` toggles immediately

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5) ALIGN (Mechanical Alignment, 6 Steps)

Use `ALIGN` after mounting/enclosing the device to remove systematic bias.

This is a guided, 6-orientation capture procedure. The device will prompt you through the positions and ask you to capture each one.

Touch Workflow

1. Tap `ALIGN`.
2. Follow the on-screen instruction (example: `Place tool: SCREEN UP`).
3. Tap `CAPTURE` to record that step.
4. Repeat until all steps are captured.
5. Tap `CANCEL` to abort safely at any time.
6. After step 6, ALIGN completes immediately and returns to normal operation.

Important:

- ALIGN and OFFSET CAL are separate operations.
- ALIGN does **not** run OFFSET CAL automatically.
- Run guided `OFFSET CAL` separately in the active `MODE` when needed.
- If you use both `SCREEN UP` and `SCREEN VERTICAL`, run `OFFSET CAL` in each mode so both mode-specific references are updated.

ACTION Button in ALIGN Workflow

If the screen is hard to access (for example screen-down steps):

- ACTION button short press = `CAPTURE`

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6) Serial Control (Optional)

If connected to a PC, you can control the same workflows via serial (115200).

Core commands:

- `z`: start guided ZERO workflow
- `a`: AXIS cycle (`BOTH -> ROLL -> PITCH`)
- `r`: ROTATE 180 toggle
- `c`: start ALIGN workflow
- `o`: start guided OFFSET CAL workflow
- `c`: context action
  - in ALIGN: `CAPTURE`
  - in ZERO pending: `CONFIRM`
  - in OFFSET CAL pending: `CONFIRM`
  - otherwise: start+confirm guided OFFSET CAL
- `y`: explicit `CONFIRM` (ZERO/OFFSET CAL workflows)
- `p`: explicit `CAPTURE` (ALIGN only)
- `m`: toggle mode immediately
- `u`: set `SCREEN UP` immediately
- `v`: set `SCREEN VERTICAL` immediately
- `x` or `n`: cancel active workflows
- `d`: toggle raw IMU debug stream (5 Hz)
- `D`: print one raw IMU sample now
- `s`: print runtime status snapshot
- `h` or `?`: print serial help
- After each serial command response, live scrolling output pauses so you can read feedback.
  - Press `Enter`, `Space`, or send `g` to resume live stream.

Serial and touch workflows are designed to stay synchronized.

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7) Web Control (Phone Remote)

Use your phone browser as a remote panel:

1. Connect to the device AP:
   - SSID: `IncidencePerfectNG-XXXX`
   - `XXXX` is derived from the board's Wi-Fi MAC identity and should differ between units.
   - Password: `incidence-ng`
2. Open `http://192.168.4.1`.

What you get in web UI:

- Live `ROLL`/`PITCH` readout with the same status line as the device.
- Battery status (`BAT/CHG`, percent, and voltage).
- Normal controls: `FREEZE`, `ZERO`, `OFFSET CAL`, `AXIS`, `MODE`, `ALIGN`, `ROTATE`, `SLEEP`.
- Context controls only when needed:
  - `CANCEL`/`CONFIRM` during guided `ZERO` and `OFFSET CAL`
  - `CANCEL`/`CAPTURE` during `ALIGN`
- Progress bar during:
  - ZERO hold/sampling
  - OFFSET CAL hold/sampling
  - ALIGN capture windows
- Expandable diagnostics panel with grouped values:
  - raw sensor vectors
  - mapped vectors
  - corrected vectors
  - physics angles + conditioning
  - calibration references (bias/zero/align)
- Network settings for:
  - Wi-Fi mode (`AP only` / `STA with AP fallback`)
  - hostname
  - battery indicator behavior
  - `Startup ZERO` on cold boot

OTA Update from Web UI

You can update firmware over Wi-Fi from the web page:

1. Open the device web UI (`http://192.168.4.1` in AP mode, or its STA IP/hostname).

2. Expand `OTA Update`.
3. Select the release-candidate firmware `.bin` built for `esp32s3`.
4. Enter the exact target firmware version shown in the release notes (`YYYY.M.X`).
5. Provide matching SHA-256:
    - use browser auto-calc if available, or
    - paste SHA-256 from release notes / local hash command below.
6. Leave `Force` unchecked for normal updates.
7. Tap `Upload & Install`.
8. Wait for reboot, reconnect, and verify the splash-screen firmware version changed to the target version.
9. If upload is rejected, correct version/SHA and retry (device keeps running current firmware when rejected).

If browser SHA-256 auto-calc is unavailable, compute hash locally and paste it:

```
Get-FileHash .\.pio\build\esp32s3\firmware.bin -Algorithm SHA256 | Select-Object -
         ExpandProperty Hash
```

If you build firmware from source, the default artifact path is:

- `.pio/build/esp32s3/firmware.bin`

Important:

- OTA updates only the firmware app image.
- If bootloader or partition layout changes, update once over USB first.
- OTA safety gates reject packages when:
    - target version is not newer than current version (unless `force` is enabled),
    - uploaded file hash does not match the provided SHA-256.
- For release testing, always verify both:
    - positive path: valid newer version + matching SHA succeeds,
    - negative path: same-version (without `force`) or wrong SHA is rejected.

Network Recovery

- In the web `Network` panel, use `Recover AP Mode` to force AP-only and clear saved STA credentials.
- Physical fallback:
    - reboot device, then immediately press+hold ACTION (`GPIO0`) continuously through startup for ~2 s
    - do not hold ACTION before power-on/reset, because `GPIO0` is a boot-strap pin
    - this applies the same AP recovery defaults (AP-only + cleared STA credentials)

Battery note:

- In web `Network` settings, `Battery` can be set to:
  - `Auto detect` (heuristic/default)
  - `Installed` (force battery-present UI semantics)
  - `No battery installed` (hide battery indicators in both web and touch UI)
- In the same `Network` settings, `Startup ZERO` can be set to:
  - `Enabled` (default; guided ZERO starts automatically on cold boot)
  - `Disabled` (keep the saved zero reference until you explicitly run ZERO)
- Current charging state is inferred from battery-voltage trend in firmware.
- The web/API exposes this via `battery_charging_inferred`.
- Battery-pack presence is best-effort on this hardware; when firmware suspects "USB powered, no pack", web/API reports `battery_present=false` with `battery_present_inferred=true`.
- On some board/charger combinations, no-pack detection may stay at `BAT/CHG` even with battery disconnected (shared rail behavior).

Web Refresh Behavior (Current)

- Live angles refresh about every `120 ms`.
- Workflow state and diagnostics refresh about every `600 ms`.
- Network status refreshes every `3000 ms`.

Important:

- Web, touch, serial, and ACTION button all share the same workflows and states.
- If one interface starts a workflow, the other interfaces should reflect it.

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8) Troubleshooting

Touch Feels Hard To Trigger

- Use ACTION button alternatives for critical actions.
- Try deliberate taps (not swipes) centered on the button.

Serial Monitor Doesn't Resume After Reset

- Some setups require closing/reopening the serial monitor after reset.
- Always report the firmware version shown on splash if you see inconsistent serial behavior.

MODE Doesn’t Apply

- MODE is immediate in current firmware.
- If orientation does not change, verify command source:
  - touch `MODE`,
  - serial `m/u/v`,
  - ACTION very-long press.

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9) Measurement Math (How It Works)

This device estimates orientation from gravity measured by the accelerometer and short-term rate from the gyro.

Accelerometer Tilt Angles

In the tool frame (`ax`, `ay`, `az`), the firmware computes:

```
roll_acc  = atan2(ay, az)
pitch_acc = atan2(-ax, sqrt(ay^2 + az^2))
```

Then values are converted to degrees.

Why `pitch` uses `sqrt(ay^2 + az^2)`:

- It uses the gravity component orthogonal to the pitch axis.
- This reduces cross-coupling from roll.
- It stays better conditioned as pitch approaches large angles.

Sensor Fusion

The live angles are fused with a complementary filter:

- Gyro integration gives smooth short-term motion response.
- Accelerometer tilt gives long-term absolute reference to gravity.

This combination is stable and responsive for field use.

Why Roll Becomes Weak Near +/-90 deg Pitch

Near pitch `+/-90 deg`, gravity aligns mostly with one axis, so roll information in the other components becomes very small. At that geometry, roll is effectively underdetermined from gravity alone.

Current firmware behavior:

- Above about `|pitch| > 80 deg`, roll is intentionally attenuated to avoid unstable swings.

Practical Guidance For Large-Angle Work

For tasks near vertical (for example checking a table-saw blade relative to the table):

1. Use the orientation/mode that makes your primary measurement axis map to `PITCH`.
2. Switch `AXIS` to `PITCH` for focus.
3. Run `OFFSET CAL` and `ZERO` in the same physical setup/orientation you will measure in.
4. Treat `ROLL` near `+/-90 deg` as secondary information.

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10) Beta Tester Checklist + Feedback

If you're testing externally, use:

- `../release/beta-checklist.md`
- `../release/tester-handoff-note.md`

When reporting an issue, include:

1. Firmware version (from splash)
2. Exact steps to reproduce
3. Expected vs actual result
4. Photos/video if UI-related

Appendix A: Hardware Notes

For reference bring-up settings (Arduino IDE), see:

- `../hardware/board-settings.md`

Authorship

This firmware and UI were developed by **Per Takman**, with assistance from **OpenAI Codex (Codex CLI)**.

License And Warranty

- The project source code is released under the MIT License.
- The software is provided “AS IS”, without warranty of any kind.
- Third-party manuals, schematics, and library dependencies remain under their respective original licenses/terms.